DO NOW — RECALL

Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

1. Binary search precondition?

2. What does low represent?

3. What is mid for bounds low0 high6?

4. Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sorted data.

2. Model answer

First possible candidate index.

3. Model answer

Index3.

4. Model answer

mid3=7, mid5=11, mid6=13, mid7=15; found index7.

By the end, you can…

• Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

• At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

• Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Vocabulary — know these terms

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

NEW LEARNING

Read, annotate and explain

Understand

Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

Trace

At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

Explain

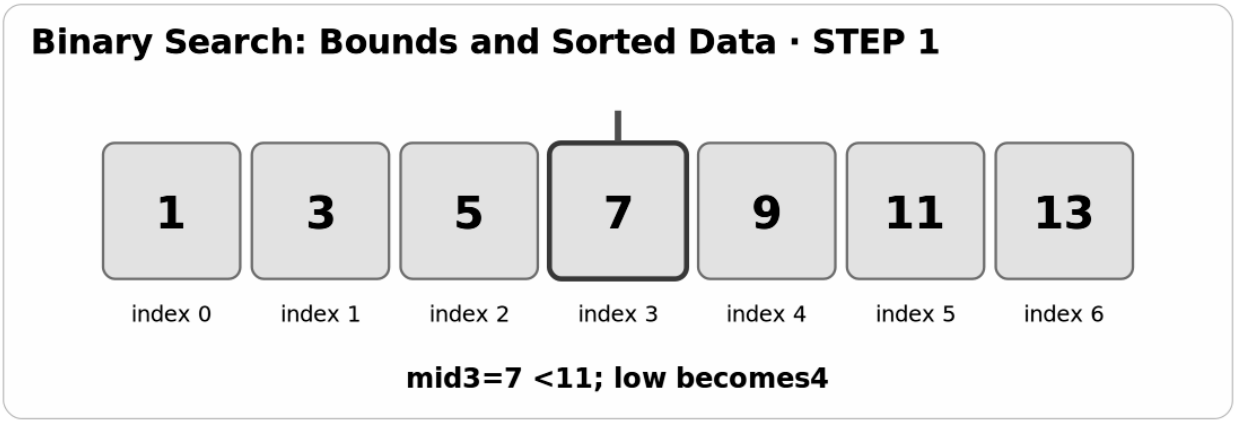
Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

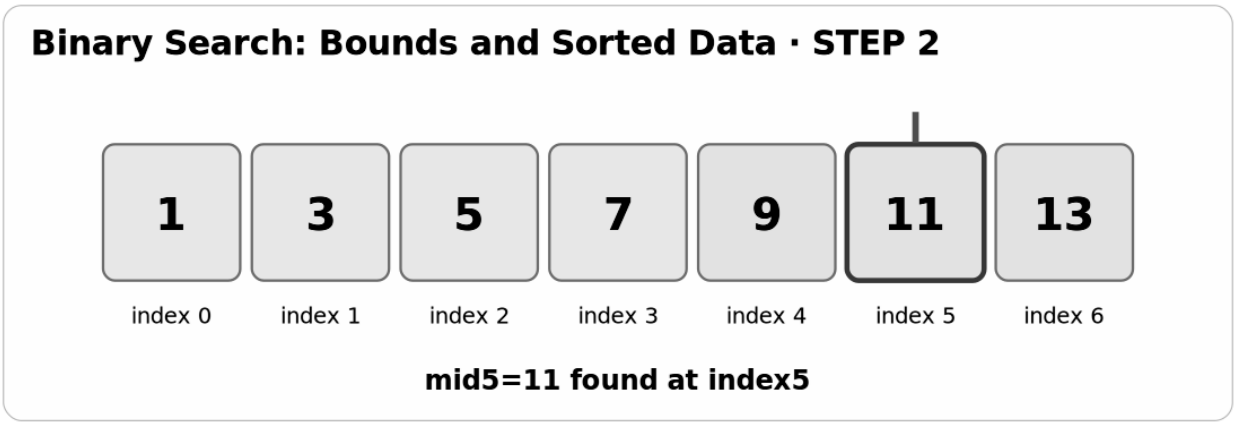
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Binary Search: Bounds and Sorted Data · STEP 1; array 1, 3, 5, 7, 9, 11, 13; mid3=7 <11; low becomes4

New Learning



Binary Search: Bounds and Sorted Data · STEP 2; array 1, 3, 5, 7, 9, 11, 13; mid5=11 found at index5

I DO — WORKED EXAMPLE

Follow the worked method

Find14 in [2,5,8,11,14,17,20].

Check the working

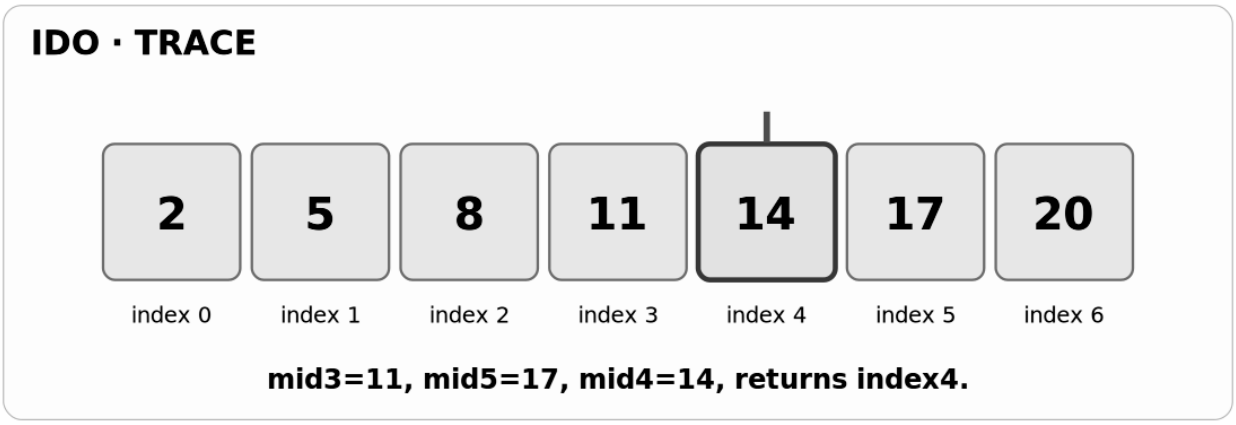
mid3=11, mid5=17, mid4=14, returns index4.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find14 in [2,5,8,11,14,17,20].

mid3=11, mid5=17, mid4=14, returns index4.



IDO · TRACE; array 2, 5, 8, 11, 14, 17, 20; mid3=11, mid5=17, mid4=14, returns index4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Find4 in [1,3,5,7,9] and explain absence.
Source code / input:
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

Question 2

Write three binary-search comparison cases and correct bounds. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

After a too-small midpoint, setting low=mid always ensures progress.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

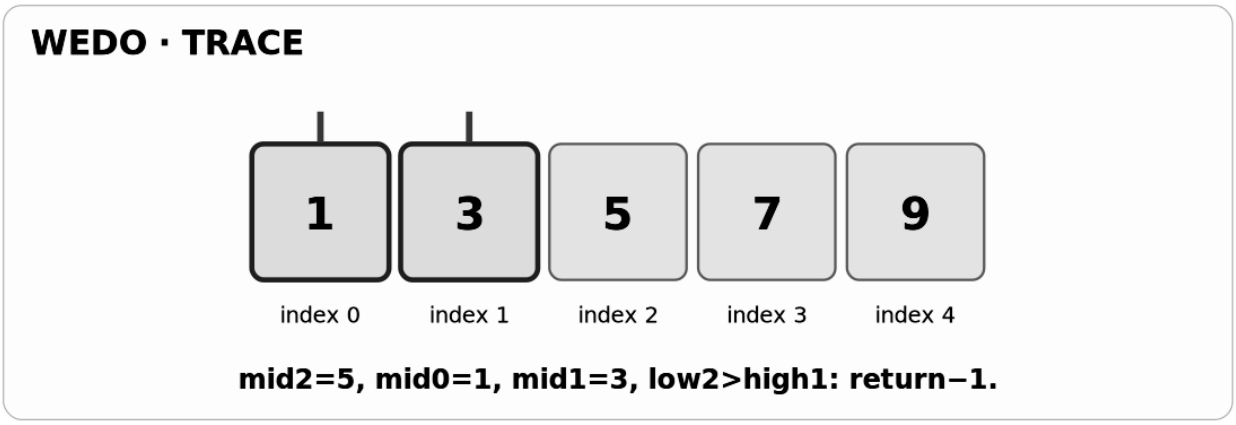
Check the working
mid2=5, mid0=1, mid1=3, low2>high1: return−1.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find4 in [1,3,5,7,9] and explain absence.

mid2=5, mid0=1, mid1=3, low2>high1: return−1.



WEDO · TRACE; array 1, 3, 5, 7, 9; mid2=5, mid0=1, mid1=3, low2>high1: return−1.